

LAB ACTIVITY 5

(Due October 28 @23:59)

Please submit the Jupiter notebook file or .py file with your answers. Please don't remove the error messages from the notebook.

1. Create a Python program that implements a random number guessing game. The program should generate a random integer between 1 and 100. The user will have to guess the number, and the program should provide feedback after each guess. The feedback should indicate whether the guessed number is too high, too low, or correct. The game should continue until the user correctly guesses the number, and it should display the number of attempts taken by the user.

Requirements:

1. Import the necessary library to generate a random number.
2. Generate a random number between 1 and 100.
3. Use a loop to allow the user to make multiple guesses.
4. After each guess, provide feedback to the user.
5. Keep track of the number of attempts taken by the user.
6. Handle invalid inputs gracefully, prompting the user to enter a valid integer if necessary.

Example Output:

```
Welcome to the Guessing Game!
I have chosen a number between 1 and 100. Try to guess it!
Enter your guess: 50
Too low! Try again.
Enter your guess: 75
Too high! Try again.
Enter your guess: 65
Congratulations! You guessed the number 65 in 3 attempts!
```

2. Write a Python program that takes the radius of a circle as input and calculates both the area and the circumference of the circle using the math module. The program should return the results and format them to two decimal places.

Requirements:

1. Import the math module.
2. Create a function called `calculate_circle_properties(radius)` that takes the radius as an argument.
3. Inside the function, calculate:
 - The function should return both the area and the circumference.
4. In the main part of the program, prompt the user to enter the radius and display the calculated area and circumference formatted to two decimal places.
 - The area and the circumference of the circle can be calculated using the formula:

- The area of the circle using the formula:

$$\text{Area} = \pi \times \text{radius}^2$$

- The circumference of the circle using the formula:

$$\text{Circumference} = 2 \times \pi \times \text{radius}$$

Example output:

```
Enter the radius of the circle: 5
The area of the circle is: 78.54
The circumference of the circle is: 31.42
```

3. Write a Python program that generates a random radius between 1 and 10 and calculates the area of a circle using that radius. The program should utilize both the random and math modules.

Requirements:

1. Import the random and math modules.
2. Create a function called `calculate_circle_area(radius)` that takes the radius as an argument and returns the area of the circle.
3. Inside the function, calculate the area of the circle
4. In the main part of the program, generate a random radius between 1 and 10 using the `random.randint()` function.
5. Call the `calculate_circle_area` function with the random radius and display the radius and the calculated area, formatted to two decimal places.

Example Output:

```
Randomly generated radius: 7  
The area of the circle is: 153.94
```